SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	2 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, IBM Watson Machine Learning

Key Features Implemented	Applicant information input through Flask web application. • Data preprocessing and feature engineering pipeline. • Credit card approval prediction using Logistic Regression, Decision Tree, Random Forest, and XGBoost. • Automatic best-model selection based on evaluation metrics. • Model serialization using Pickle/Joblib. • Real-time prediction through Flask web interface. • IBM Watson Machine Learning cloud deployment. • Input validation and error handling.
Pending / Incomplete Features	User authentication and role-based access control, prediction history storage in a database, email/SMS notification for prediction results, and advanced analytics dashboard.
Setup / Run Instructions	1. Install Python 3.8 or above. 2. Install required packages using pip install -r requirements.txt. 3. Run the model training notebook or use the saved model file. 4. Start the Flask application using python app.py. 5. Open http://127.0.0.1:5000 in a web browser. 6. Enter applicant details and click **Predict** to obtain the approval result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture by separating data preprocessing, model training, prediction, and web application components.

Multiple machine learning algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost) were evaluated to identify the best-performing model.

The trained model is integrated into a Flask web application for real-time credit card approval prediction.
IBM Watson Machine Learning deployment enables cloud-based access and scalability.

The solution is extensible and can be enhanced by integrating a database, authentication, audit logging, and advanced reporting features.